

# SUSHANT KUMAR

sushant.eel1iitd@gmail.com | +91 84331 90412 | linkedin.com/in/sushant-kumar-567b65253 | sushantxps.github.io/portfolio | New Delhi, India

## EDUCATION

### Indian Institute of Technology, Delhi

2022 to 2026

B.Tech in Electrical Engineering | Dept. change Mech. to Elec. Engg., 2023

- Relevant coursework: Machine Intelligence & Learning, Statistical Signal Processing, Probability & Stochastic Processes, Image Processing, Optimisation Methods, Linear Algebra, Computer Architecture.

## EXPERIENCE

### Computer Vision Intern, Autonomous Drone Navigation

Jul to Nov 2024

Botlab Dynamics Pvt. Ltd., Unmanned Aerial Systems Division

- Built the drone perception pipeline around a calibrated stereo pair, converting disparity maps into metric depth to drive obstacle detection and feed the navigation stack.
- Integrated a visual inertial odometry and SLAM stack for real time 6 DoF pose: handled camera and IMU data ingestion, coordinate frame transformations between sensor, body and world frames, and validated localisation performance in live flight rather than on logged data alone.
- Deployed to an NVIDIA Jetson platform and optimised inference by converting models to TensorRT with FP16 quantisation, reaching real time throughput inside the platform's compute and power envelope.
- Estimated and validated intrinsic and extrinsic camera parameters using standard calibration procedures, and verified camera and IMU synchronisation by checking timestamps and sensor alignment, so that perception and localisation consumed temporally consistent data.
- Refined preprocessing for robust detection under noisy, low light flight; analysed flight test telemetry to diagnose where perception failed in the field rather than on benchmarks.

## RESEARCH EXPERIENCE

### Coauthor, Hyperbolic Geometry of Reasoning in LLM Hidden States

2025

Submitted to Mechanistic Interpretability Workshop @ ICML 2026

- Implemented the Poincaré ball probe architecture: spectrally normalised projection, learnable bounded scaling, exponential map, and Maximum Distance Rescaling for gradient stability.
- Probe recovered reasoning depth from the hidden states of a 7B model (DeepSeek-R1-Distill-Qwen-7B) at Spearman  $\rho = 0.967$ , while the matched Euclidean probe collapsed to  $\rho = 0.488$ . The gap is the result: the representation carries hierarchical structure a flat geometry cannot express. Tuned curvature and embedding dimension across systematic ablations in reproducible PyTorch pipelines.

### Research Intern to B.Tech Project, Efficient Video Perception under Compute Budget

May 2025 to present

Neuro Computing Lab, IIT Delhi | Prof. Tapan Kumar Gandhi & Dr. Manvir Bhatia

- Detection pipeline for subpixel involuntary movement in long duration clinical video, a weak signal recovery problem where the target motion sits near the sensor noise floor: denoised optical flow, adaptive temporal smoothing, dynamic thresholding. 40%+ improvement in subtle motion detection over baseline, validated against ground truth annotated by clinicians.
- Compressing the optical flow stage to run within a low power embedded budget, trading dense flow computation against detection sensitivity to keep accuracy under a fixed compute ceiling.

### Research Contributor, RTLGen Project (Remote)

Jan to Jul 2024

Edge Computing Lab, Harvard University | Prof. Vijay Janapa Reddi

- Built LangChain pipelines for automated prompt feedback evaluation of generated hardware designs within the RTLGen Bench Harness, at the intersection of ML and computer architecture. Acknowledged in the resulting publication (arXiv:2501.01892).

## SELECTED PROJECTS

### Quantum Inspired Neural Network for Vision and Brain Decoding

May 2025 to present

Prof. Tapan K. Gandhi, Neuro Computing Lab, IIT Delhi. Exploratory research into quantum inspired representations for decoding semantic structure from brain imaging signals recorded during human visual perception.

### Handwritten Script Classification, Neural Network from Scratch and CNN

Sep to Oct 2024

Prof. Rahul Garg, Dept. of Computer Science, IIT Delhi. Implemented a five layer feedforward network from scratch in NumPy with custom backpropagation; built a six layer PyTorch CNN reaching 90% test accuracy in 8 epochs.

## TECHNICAL SKILLS

**Robotics perception:** stereo depth estimation, disparity to metric depth, visual inertial odometry, SLAM integration, camera intrinsic and extrinsic calibration, camera and IMU synchronisation, coordinate frame transformations, optical flow, object tracking, obstacle detection.

**Onboard and edge deployment:** NVIDIA Jetson, TensorRT, FP16 quantisation, real time inference under compute and power budget, model compression, latency profiling.

**Machine learning:** PyTorch, TensorFlow, scikit-learn, NumPy, pandas, SciPy; transformers, CNNs, RNNs, representation probing, distillation, ablation design.

**Programming and tooling:** Python, C++, C, MATLAB, Java, TypeScript, SQL, Verilog; ROS2, Git, Docker, LaTeX, Jupyter, LangChain, PyTest.

**Mathematics:** probability and stochastic processes, statistical signal processing, linear algebra, convex optimisation.

## ACHIEVEMENTS

- **JEE Advanced 2022:** Rank 417 (top 0.2%) in India's most competitive engineering entrance. Indian Mathematics Olympiad Qualifier (2020), UP region.
- **Lead Organiser, Yann LeCun Panel at IIT Delhi (Oct 2024):** led a three person team for a 5,000 attendee event with Meta's Chief AI Scientist; primary liaison with Meta's global communications team.